Some New Results on System Identification with Dynamic Neural Networks

Wen Yu and Xiaoou Li

***Abstract*—Nonlinear system on-line identification via dynamic neural networks is studied in this paper. The main contribution of the paper is that the passivity approach is applied to access several new stable properties of neuro identification. The conditions for passivity, stability, asymptotic stability, and input-to-state stability are established in certain senses. We conclude that the gradient descent algorithm for weight adjustment is stable in an L_∞ sense and robust to any bounded uncertainties.**

***Index Terms*—Dynamic neural networks, passivity, system identification.**

I. Introduction

RECENT results show that neural-network techniques seem to be very effective to identify a wide class of complex nonlinear systems when we have no complete model information, or even when we consider the controlled plant as a black box [3]. This model-free approach uses the nice feature of neural networks, but the lack of model makes hard to obtain theoretical result on the stability of neuro identification. For the engineers it is very important to assure the stability in theory before they want to apply neural networks to real systems.

In spite of their successful applications, there are not many results on the stability analysis for neural networks. The global asymptotic stability (GAS) of dynamic neural networks has been developed during the last decade. Diagonal stability [6] and negative semidefiniteness [7] of the interconnection matrix may make Hopfield–Tank neuro circuit GAS. Multilayer perceptrons (MLP) and recurrent neural networks can be related to the Lur'e systems, the absolute stabilities were developed by [14] and [9]. Input-to-state stability (ISS) technique [13] is an effective tool for dynamic neural networks, [16] stated that if the weights are small enough, neural networks are ISS and GAS with zero input. Many publications investigated the stability of identification and tracking errors of neural networks. Reference [5] studied the stability condition when multilayer perceptrons were used to identify and control nonlinear systems. Lyapunov-like analysis is suitable for dynamic neural networks. References [12] and [17] discussed the signal-layer case. References [8] and [10] studied the high-order dynamic networks and multilayer dynamic networks. Since neural networks cannot match nonlinear systems exactly, some robust modifications [4] were applied on the normal gradient descent algorithm [12], [14] and the backpropagation algorithm [5], [17].

Manuscript received April 17, 2000; revised October 23, 2000.

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Publisher Item Identifier S 1045-9227(01)02039-2.

It is well known that neuro identification is in sense of black-box identification. All of the uncertainties can be considered as parts of the black-box. Hence, the commonly used robustifying techniques such as dead-zone or σ-modification may be not necessary. This is the motivation of this paper.

We will use passivity theory to analyze the stability of neuro identification. Passivity approach may deal with the poor defined nonlinear systems, and offers elegant solutions for the proof of absolute stability. It can lead to general conclusions on the stability by using only input–output characteristics. The passivity properties of MLP were examined in [2]. By means of analyzing the interconnected of error models, they derived the relationship between passivity and closed-loop stability. To the best of our knowledge, open-loop analysis based on the passivity method for dynamic neural networks has not yet been established in the literature. We will show that a gradient-like learning law can make the identification error stability, asymptotic stability and input-to-state stability. A Simulations of the vehicle idle speed identification gives the effectiveness of the proposed algorithm.

II. Preliminaries

Consider a class of nonlinear systems described by

$$\dot{x}_t = f(x_t, u_t), \qquad y_t = h(x_t, u_t) \tag{1}$$

where

$x_t \in \Re^n$ state;
$u_t \in \Re^m$ input vector;
$y_t \in \Re^m$ output vector.

f: $\Re^n \times \Re^m \to \Re^n$ is locally Lipschitz, h: $\Re^n \times \Re^m \to \Re^m$ is continuous. It is also assumed that for any $x^0 = x_0 \in \Re^n$, the output $y_t = h(\Phi(t, x^0, u))$ of (1) is such that $\int_0^t |u_s^T y_s|\, ds < \infty$, for all $t \geq 0$, i.e., the energy stored in system (1) is bounded. Following to [1], let us now recall some passivity properties as well as some stability properties of passive systems.

Definition 1: A system (1) is said to be *passive* if there exists a C^r nonnegative function $S(x_t)$: $\Re^n \to \Re$, called *storage function*, such that, for all u_t, all initial conditions x^0 and all $t \geq 0$ the following inequality holds:

$$\dot{S}(x_t) \leq u_t^T y_t - \varepsilon u_t^T u_t - \delta y_t^T y_t - \rho\psi(x_t)$$
$$(x_t, u_t) \in \Re^n \times \Re^m$$

where ε, δ and ρ are nonnegative constants, $\psi(x_t)$ is positive semidefinite function of x_t such that $\psi(0) = 0$. $\rho\psi(x_t)$ is called state dissipation rate. Furthermore, the system is said to be *strictly passive* if there exists a positive definite function $V(x_t)$: $\Re^n \to \Re$ such that $\dot{S}(x_t) \leq u_t^T y_t - V(x_t)$.

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Property 1: If the storage function $S(x_t)$ is differentiable and the dynamic system is passive, storage function $S(x_t)$ satisfies $\dot{S}(x_t) \le u_t^T y_t$.

Definition 2: A system (1) is said to be globally *input-to-state stability* if there exists a $\mathcal{K}$-function $\gamma(s)$ [continuous and strictly increasing $\gamma(0) = 0$] and $\mathcal{KL}$-function $\beta(s, t)$ [$\mathcal{K}$-function and for each fixed $s_0 \ge 0$, $\lim_{t\to\infty} \beta(s_0, t) = 0$], such that, for each $u \in L_\infty$ ($\sup\{\|u(t)\|, t \ge 0\} < \infty$) and each initial state $x^0 \in R^n$, it holds that

$$\|x(t, x^0, u_t)\| \le \beta(s, t)(\|x^0\|, t) + \gamma(s)\|u_t\|$$

for each $t \ge 0$.

Property 2: If a system is input-to-state stability, the behavior of the system should remain bounded when its inputs are bounded.

III. Neuro Identification via Passivity Technique

The identified nonlinear system is given by

$$\dot{x}_t = f(x_t, u_t, t), \qquad x_t \in \Re^n, \quad u_t \in \Re^m. \tag{2}$$

We construct following dynamic neural network for identification

$$\dot{\hat{x}}_t = A\hat{x}_t + W_{1,t}\sigma(\hat{x}_t) + W_{2,t}\phi(\hat{x}_t)\gamma(u_t) \tag{3}$$

where $\hat{x}_t \in \Re^n$ is the state of the neural network, $A \in \Re^{n\times n}$ is a stable matrix. $W_{1,t} \in \Re^{n\times n}$, $W_{2,t} \in \Re^{n\times n}$ are weight matrices of neural networks. The vector functions $\sigma(x_t) \in \Re^n$ is assumed to be n-dimensional with the elements increasing monotonically. The matrix function $\phi(\cdot)$ is assumed to be $\Re^{n\times m}$ diagonal: $\phi(\hat{x}_t) = \mathrm{diag}(\phi_1(\hat{x}_1)\cdots\phi_n(\hat{x}_n))$. $\gamma(u_t) \in \Re^m$. Function $\gamma(\cdot)$ is selected as $\|\gamma(u_t)\|^2 \le \overline{u}$. The typical presentation of the elements $\sigma_i(\cdot)$ and $\phi_i(\cdot)$ are as sigmoid functions, i.e.,

$$\sigma_i(x_i) = \frac{a_i}{1 + e^{-b_i x_i}} - c_i.$$

Remark 1: The dynamic neural network (3) has been discussed by many authors, for example, [12], [8], [10] and [17]. It can be seen that Hopfield model is the special case of this networks with $A = \mathrm{diag}\{a_i\}$, $a_i := -1/R_iC_i$, $R_i > 0$ and $C_i > 0$. R_i and C_i are the resistance and capacitance at the ith node of the network, respectively.

Let us define identification error as

$$\Delta_t = \hat{x}_t - x_t.$$

Because $\sigma(\cdot)$ and $\phi(\cdot)$ are chosen as sigmoid functions, clearly they satisfy the following *Lipshitz* property:

$$\begin{aligned} \tilde{\sigma}^T\Lambda_1\tilde{\sigma} &\le \Delta_t^T D_\sigma \Delta_t \\ (\tilde{\phi}_t\gamma(u_t))^T\Lambda_2(\tilde{\phi}_t\gamma(u_t)) &\le \overline{u}\Delta_t^T D_\phi \Delta_t \end{aligned} \tag{4}$$

where $\tilde{W}_{1,t} = W_{1,t} - W_1^*$, $\tilde{W}_{2,t} = W_{2,t} - W_2^*$, $\tilde{\sigma} = \sigma(\hat{x}_t) - \sigma(x_t)$, $\tilde{\phi} = \phi(\hat{x}_t) - \phi(x_t)$, Λ_1, Λ_2, D_σ and D_ϕ are known positive constants, nonlinear.

Generally, dynamic neural network (3) cannot follow the nonlinear system (2) exactly. The nonlinear system may be written as

$$\dot{x}_t = Ax_t + W_1^*\sigma(x_t) + W_2^*\phi(x_t)\gamma(u_t) - \tilde{f}_t \tag{5}$$

where W_1^* and W_2^* are bounded unknown matrices

$$W_1^*\Lambda_1^{-1}W_1^{*T} \le \overline{W}_1, \qquad W_2^*\Lambda_2^{-1}W_2^{*T} \le \overline{W}_2. \tag{6}$$

$\overline{W}_1$ and $\overline{W}_2$ are prior known matrices, vector function $\tilde{f}_t$ can be regarded as modeling error and disturbances. The multilayer dynamics is obtained from (3) and (5)

$$\begin{aligned} \dot{\Delta}_t = {} & A\Delta_t + \tilde{W}_{1,t}\sigma(\hat{x}_t) + \tilde{W}_{2,t}\phi(\hat{x}_t)\gamma(u_t) \\ & + W_1^*\tilde{\sigma} + W_1^*\tilde{\phi}\gamma(u_t) + \tilde{f}_t. \end{aligned} \tag{7}$$

If we define

$$R = \overline{W}_1 + \overline{W}_2, \qquad Q = D_\sigma + \overline{u}D_\phi + Q_0 \tag{8}$$

and the matrices A and Q_0 are selected to fulfill the following conditions:

1) the pair $(A, R^{1/2})$ is controllable, the pair $(Q^{1/2}, A)$ is observable;
2) local frequency condition satisfies

$$A^TR^{-1}A - Q \ge \tfrac{1}{4}[A^TR^{-1} - R^{-1}A]R[A^TR^{-1} - R^{-1}A]^T \tag{9}$$

then the following assumption can be established.

A1: There exist a stable matrix A and a strictly positive definite matrix Q_0 such that the matrix Riccati equation

$$A^TP + PA + PRP + Q = 0 \tag{10}$$

has a positive solution $P = P^T > 0$.

This conditions is easily fulfilled if we select A as stable diagonal matrix. The next theorem states the learning procedure of neuro identifier.

Theorem 1: Subject to assumption A1 being satisfied, if the weights $W_{1,t}$ and $W_{2,t}$ are updated as

$$\begin{aligned} \dot{W}_{1,t} &= -K_1P\Delta_t\sigma^T(\hat{x}_t) \\ \dot{W}_{2,t} &= -K_2P\phi(\hat{x}_t)\gamma(u_t)\Delta_t^T \end{aligned} \tag{11}$$

where P is the solution of the Riccati equation (10), then the identification error dynamics (7) is strictly passive from the modeling error $\tilde{f}_t$ to the identification error $2P\Delta_t$.

Proof: Select a Lyapunov function (storage function) as

$$S_t = \Delta_t^TP\Delta_t + \mathrm{tr}\{\tilde{W}_{1,t}^TK_1^{-1}\tilde{W}_{1,t}\} + \mathrm{tr}\{\tilde{W}_{2,t}^TK_2^{-1}\tilde{W}_{2,t}\} \tag{12}$$

where $P \in \Re^{n\times n}$ is positive definite matrix. According to (7), the derivative is

$$\begin{aligned} \dot{S}_t = {} & \Delta_t^T(PA + A^TP)\Delta_t + 2\Delta_t^TP\tilde{W}_{1,t}\sigma(\hat{x}_t) \\ & + 2\Delta_t^TP\tilde{W}_{2,t}\phi(\hat{x}_t)\gamma(u_t) + 2\Delta_t^TP\tilde{f}_t \\ & + 2\Delta_t^TP[W_1^*\tilde{\sigma} + W_1^*\tilde{\phi}\gamma(u_t)] + 2\mathrm{tr}\{\dot{\tilde{W}}_{1,t}^TK_1^{-1}\tilde{W}_{1,t}\} \\ & + 2\mathrm{tr}\{\dot{\tilde{W}}_{2,t}^TK_2^{-1}\tilde{W}_{2,t}\}. \end{aligned}$$

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Since $\Delta_t^T P W_1^* \tilde{\sigma}_t$ is scalar, using (4) and the following matrix inequality:

$$X^T Y + (X^T Y)^T \le X^T \Lambda^{-1} X + Y^T \Lambda Y \tag{13}$$

where X, Y, $\Lambda \in \Re^{n\times k}$ are any matrices, Λ is any positive definite matrix, we obtain

$$\begin{aligned} 2\Delta_t^T P W_1^* \tilde{\sigma}_t &\le \Delta_t^T P W_1^* \Lambda_1^{-1} W_1^{*T} P \Delta_t + \tilde{\sigma}_t^T \Lambda_1 \tilde{\sigma}_t \\ &\le \Delta_t^T (P\overline{W}_1 P + D_\sigma)\Delta_t \\ 2\Delta_t^T P W_2^* \tilde{\phi}_t \gamma(u_t) &\le \Delta_t^T (P\overline{W}_2 P + \overline{u} D_\phi)\Delta_t. \end{aligned} \tag{14}$$

So we have

$$\begin{aligned} \dot{S}_t \le \Delta_t^T [&PA + A^T P + P(\overline{W}_1 + \overline{W}_2)P \\ &+ (D_\sigma + \overline{u} D_\phi + Q_0)]\Delta_t + 2\mathrm{tr}\left\{ \dot{\tilde{W}}_{1,t}^T K_1^{-1} \tilde{W}_{1,t} \right\} \\ &+ 2\Delta_t^T P \tilde{W}_{1,t} \sigma(\hat{x}_t) + 2\Delta_t^T P \tilde{f}_t + 2\mathrm{tr}\left\{ \dot{\tilde{W}}_{2,t}^T K_2^{-1} \tilde{W}_{2,t} \right\} \\ &+ 2\Delta_t^T P \tilde{W}_{2,t} \phi(\hat{x}_t) \gamma(u_t) - \Delta_t^T Q_0 \Delta_t. \end{aligned}$$

Since $\dot{\tilde{W}}_{1,t} = \dot{W}_{1,t}$, if we use the updating law as in (11) and *A1*, we have

$$\dot{S}_t \le -\Delta_t^T Q_0 \Delta_t + 2\Delta_t^T P \tilde{f}_t. \tag{15}$$

From Definition 1, if we define the input as $\tilde{f}_t$ and the output as $2P\Delta_t$, then the system is strictly passive with $V_t = \Delta_t^T Q_0 \Delta_t \ge 0$. ∎

Remark 2: Since the updating rate is $K_i P$ ($i = 1, 2$), and K_i can be selected as any positive matrix, the learning process of the dynamic neural network (11) is free of the solution of the Riccati equation (10).

Corollary 1: If only parameters uncertainty present ($\tilde{f}_t = 0$), then the updating law as (11) can make the identification error asymptotic stable

$$\begin{aligned} &\lim_{t\to\infty} \Delta_t = 0, \qquad W_{1,t} \in L_\infty, \qquad W_{2,t} \in L_\infty \\ &\lim_{t\to\infty} \dot{W}_{1,t} = 0, \qquad \lim_{t\to\infty} \dot{W}_{2,t} = 0. \end{aligned} \tag{16}$$

Proof: Since the identification error dynamics (7) is passive and $\tilde{f}_t = 0$, from the *Property 1* the storage function $S(x_t)$ satisfies

$$\dot{S}(x_t) \le \tilde{f}_t^T 2P\Delta_t = 0.$$

The positive definite $S(x_t)$ implies Δ_t, $W_{1,t}$ and $W_{2,t}$ are bounded. From the error equation (7) $\dot{\Delta}_t \in L_\infty$

$$\dot{S} \le -\Delta_t^T Q_0 \Delta_t \le 0.$$

Integrate (15) both sides

$$\int_0^\infty \|\Delta_t\|_{Q_0} \le S_0 - S_\infty < \infty.$$

So $\Delta_t \in L_2 \cap L_\infty$, using Barlalat's Lemma [4] we have (16). Since u_t, $\sigma(\hat{x}_t)$, $\phi(\hat{x}_t)$ and P are bounded, $\lim_{t\to\infty} \dot{W}_{1,t} = 0$ and $\lim_{t\to\infty} \dot{W}_{1,t} = 0$. ∎

Remark 3: For model matching case, Lyapunov-like analysis can reach the same result (see [17, Corollaries 1 and 2]). But in present of modeling error, we will give an absolutely different result: the gradient descent algorithm (11) is also robust respected to the bounded unmodeled dynamic and disturbance.

Theorem 2: If assumption A1 is satisfied and there exists a positive definite matrix Λ_f such that

$$\lambda_{\max}(P\Lambda_f P) \le \lambda_{\min}(Q_0) \tag{17}$$

then the updating law (11) may make the dynamics of neuro identifier (7) input-to-state stability (ISS).

Proof: In view of the matrix inequality (13)

$$2\Delta_t^T P \tilde{f}_t \le \Delta_t^T P \Lambda_f P \Delta_t + \tilde{f}_t^T \Lambda_f^{-1} \tilde{f}_t. \tag{18}$$

Equation (15) can be represented as

$$\begin{aligned} \dot{S}_t &\le -\Delta_t^T Q_0 \Delta_t + 2\Delta_t^T P \tilde{f}_t \\ &\le -\lambda_{\min}(Q_0)\|\Delta_t\|^2 + \Delta_t^T P \Lambda_f P \Delta_t + \tilde{f}_t^T \Lambda_f^{-1} \tilde{f}_t \\ &\le -\alpha_{\|\Delta_t\|}\|\Delta_t\| + \beta_{\|\tilde{f}_t\|}\|\tilde{f}_t\| \end{aligned}$$

where $\alpha_{\|\Delta_t\|} := [\lambda_{\min}(Q_0) - \lambda_{\max}(P\Lambda_f P)]\|\Delta_t\|$, $\beta_{\|\tilde{f}_t\|} := \lambda_{\max}(\Lambda_f^{-1})\|\tilde{f}_t\|$. We can select a positive definite matrix Λ_f such that (17) is established. So α and β are $\mathcal{K}_\infty$ functions, S_t is an ISS-Lyapunov function. Using [13, Th. 1], the dynamic of identification error (7) is input to state stability. ∎

Remark 4: First, we may choose A and Q_0 such that the Riccati equation (10) has positive solution P. Then Λ_f may be found according to the condition (17). Since (18) is correct for any positive definite matrix, (17) can be established if Λ_f is selected as a small enough constant matrix. The condition (17) has no effect on the network dynamics (3) and its training (11).

Corollary 2: If the modeling error $\tilde{f}_t$ is bounded, then the updating law (11) can make the identification procedure stable

$$\Delta_t \in L_\infty, \qquad W_{1,t} \in L_\infty, \qquad W_{2,t} \in L_\infty.$$

Proof: From Property 2 we know input-to-state stability means that the behavior of the dynamic neural networks should remain bounded when its input is bounded. ∎

Remark 5: Since the state and output variables are physically bounded, the modeling error $\tilde{f}_t$ can be assumed to be bounded too (see, for example, [5], [10], [12]). Unlike robust adaptive laws, such as dead-zone and σ-modification [8], the upper bound of modeling error is not needed for the network training.

Remark 6: It is well known that structure uncertainties will cause the parameters drift for adaptive control, so one has to use robust modification to make identification error stability [4]. Robust adaptive methods may be extended to neuro identification directly [5], [10], [12]. But neuro identification is a kind of black-box method, structure information is useless and all of uncertainties are inside the black-box. Although robust adaptive algorithms are suitable for neuro identification, they are not the simplest. By means of passivity technique, we success to prove our conclusion: the gradient descent algorithm for the weights is robust with respect to any bounded uncertainties.

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IV. Simulation

The engine operation at idle is a nonlinear process that is far from its optimal range. Because it does not require any large degree of instrumentation or external sensing capabilities, the idle speed control is also accessible and can be formatted as a benchmark problem for control society. The process of engine at idle has time delays that vary inversely with engine speed and is time-varying due to aging of components and environmental changes such as engine warm-up after a cold start. The measurement of system outputs occurs asynchronously with the calculation of control signals. We assume that the occurrence of plant disturbances, such as engagement of air conditioner compressor, shift from neutral to drive in automatic transmissions, application, and release of electric loads, and power steering lock-up, are not directly measured. The dynamic engine model is a two inputs and two outputs system [11]

$$\begin{aligned}\dot{P} &= k_P(\dot{m}_{ai} - \dot{m}_{ao}), \qquad \dot{N} = k_N(T_i - T_L)\\ \dot{m}_{ai} &= (1 + k_{m1}\theta + k_{m2}\theta^2)g(P),\\ \dot{m}_{ao} &= -k_{m3}N - k_{m4}P + k_{m5}NP + k_{m6}NP^2.\end{aligned}$$

The parameters for an engine model with 1.6-L four-cylinder fuel injected are

$$\begin{aligned}g(P) &= \begin{cases}1 & P < 50.6625\\ 0.0197\sqrt{101.325P - P^2} & P \geq 50.6625\end{cases}\\ T_i &= -39.22 + 325\,024 m_{ao} - 0.0112\delta^2 + 0.635\delta\\ &\quad + \frac{2\pi}{60}(0.0216 + 0.000\,675\delta)N - \left(\frac{2\pi}{60}\right)^2 0.000\,102N^2\\ T_L &= \left(\frac{N}{263.17}\right)^2 + T_d\\ m_{ao} &= \dot{m}_{ao}(t-\tau)/(120N)\end{aligned}$$

where

$k_P = 42.40$;
$k_N = 54.26$;
$k_{m1} = 0.907$;
$k_{m2} = 0.0998$;
$k_{m3} = 0.000\,596\,8$;
$k_{m4} = 0.000\,534\,1$;
$k_{m5} = 0.000\,001\,757$;
$\tau = 45/N$.

The system outputs are manifold press P (kPa) and engine speed N (r/min). The control inputs are throttle angle θ (degree) and the spark advance δ (degree). Disturbances act to the engine in the form of unmeasured accessory torque T_d (N-m). The variable $\dot{m}_{ai}$ and $\dot{m}_{ao}$ refer to the mass air flow into and out of the manifold. m_{ao} is the air mass in the cylinder. The parameter τ is a dynamic transport time delay. The function $g(P)$ is a manifold pressure influence function. T_i is the engine's internally developed torque, T_L is the load torque. If we define $x = (P, N)^T$, $u = (\theta, \delta)^T$, then the dynamic of vehicle idle speed are

$$\dot{x}_t = \begin{pmatrix}\dot{x}_1\\ \dot{x}_2\end{pmatrix} = \begin{pmatrix}f_1(x, u)\\ f_2(x, u)\end{pmatrix}$$

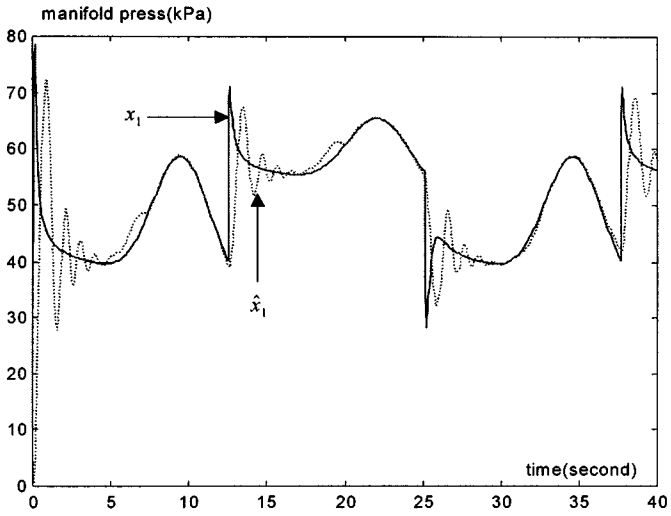


Fig. 1. Neuroidentification for x_1 (gradient).

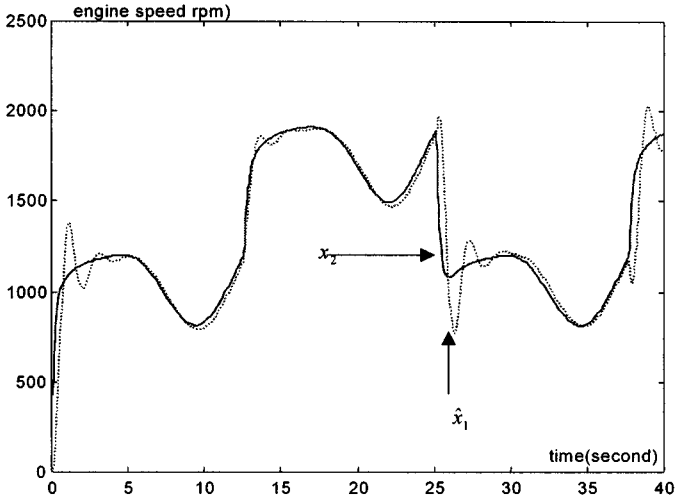


Fig. 2. Neuroidentification for x_2 (gradient).

where f_1 and f_2 are assumed to be unknown and only x and u are measurable. In order to do the simulation, we select input as $\delta = 30\sin(t/2)$, θ is sawtooth wave with amplitude ten, frequency 1/2, T_d is square wave with amplitude 20, frequency 1/4, $x_0 = [10, 500]^T$. Let us select dynamic neural network as

$$\dot{\hat{x}}_t = A\hat{x}_t + W_{1,t}\sigma(\hat{x}_t) + W_{2,t}\phi(\hat{x}_t)u_t, \quad W_{1,t}, W_{2,t} \in R^{2\times 2}.$$

The sigmoid functions are $\sigma(x_i) = 2/(1+e^{-2x_i}) - 0.5$, $\phi(x_i) = 0.2/(1+e^{-0.2x_i}) - 0.05$. We also select $A = \left[\begin{smallmatrix}-2 & 0\\ 0 & -2\end{smallmatrix}\right]$, $K_1P = K_2P = 2I$, $\hat{x}_0 = [-5, -5]^T$. We use the learning law

$$\begin{aligned}\dot{W}_{1,t} &= -K_1P\Delta_t\sigma(\hat{x}_t)^T\\ \dot{W}_{2,t} &= -K_2P\phi(\hat{x}_t)^T u_t\Delta_t^T\end{aligned} \tag{19}$$

$W_{1,0} = \left[\begin{smallmatrix}1 & 10\\ 10 & 1\end{smallmatrix}\right]$, $W_{2,0} = \left[\begin{smallmatrix}0.1 & 0\\ 0 & 0.1\end{smallmatrix}\right]$. The following simulation results show the identification ability of dynamic neural network. Figs. 1 and 2 are with internal uncertainties (unmodeled dynamics and accessory torque). We may find that although there

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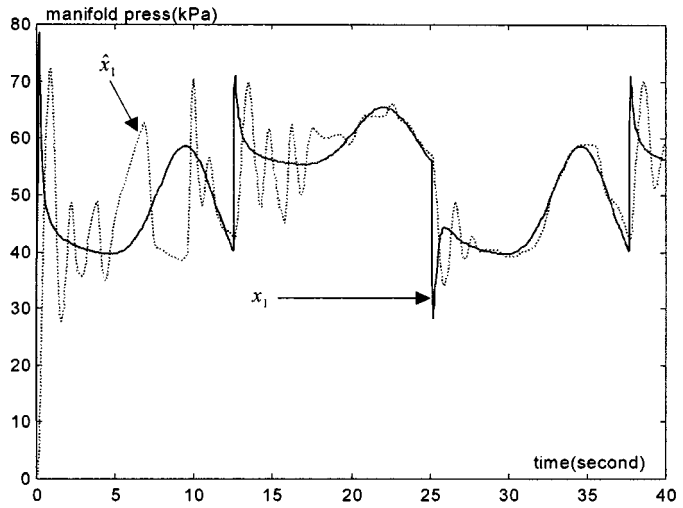


Fig. 3. Neuroidentification for x_1 (dead-zone).

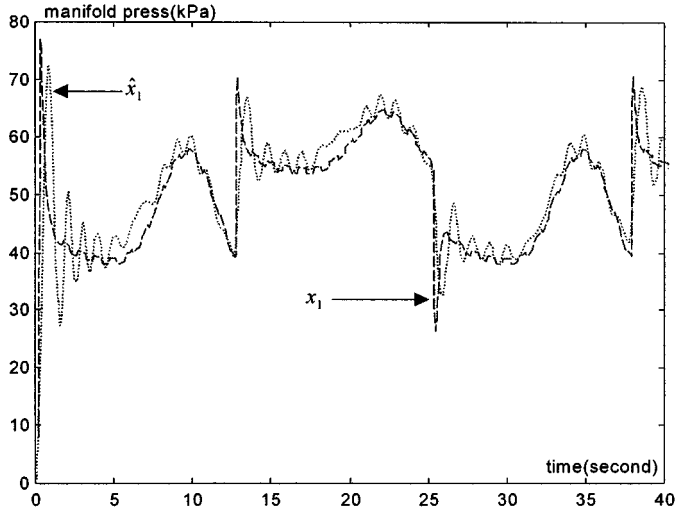


Fig. 4. Neuroidentification for x_1 (gradient).

exists modeling error, gradient descent algorithm (19) is stable. If we define the relative identification error as

$$\overline{e} = \frac{\int \Delta_t \, dt}{\int x_t \, dt}.$$

$\overline{e}_1$ corresponding to Fig. 1 is 7.40%, and $\overline{e}_2$ corresponding to Fig. 2 is 4.73%. If we use robust modification algorithm, for example dead-zone [17], the upper bound of the modeling error is selected as ten, the identification result is shown in Fig. 3. The relative identification error $\overline{e}_1$ corresponding to Fig. 3 is 12.05%. The identification accuracy of (19) is better than robust modification algorithm [17].

Now we put external uncertainties on the plant. We introduce two square waves in the measurement outputs, the frequencies are 1 Hz, the amplitudes are 0.5 and 10. Figs. 4 and 5 shown the identification results with bounded disturbances. The relative identification errors are $\overline{e}_1 = 7.81\%$, $\overline{e}_2 = 4.67\%$. One can see

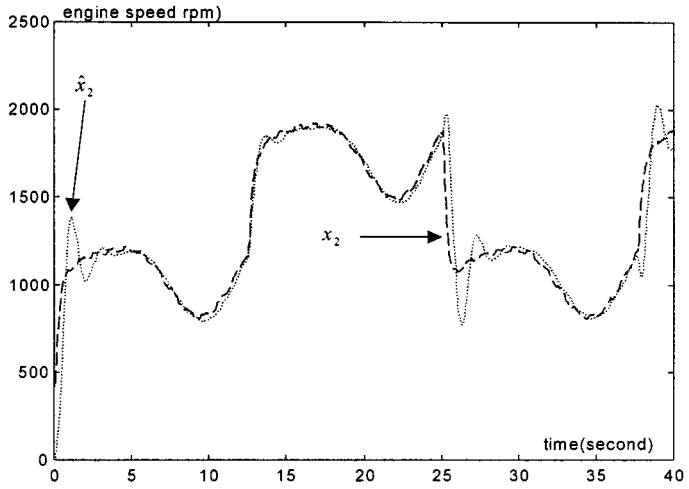


Fig. 5. Neuroidentification for x_2 (gradient).

that the gradient algorithm of dynamic neural network is also robust with respect to bounded disturbances.

V. Conclusion

In this paper, the passivity approach is used to prove that a gradient descent algorithm for weight adjustment is stable and robust to any bounded uncertainties, including the optimal network approximation error. Hence we conclude that the commonly used robustifying techniques such as dead-zone and σ-modification are not necessary.

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